DTC P0299:00 [SKYACTIV-D 1.5]

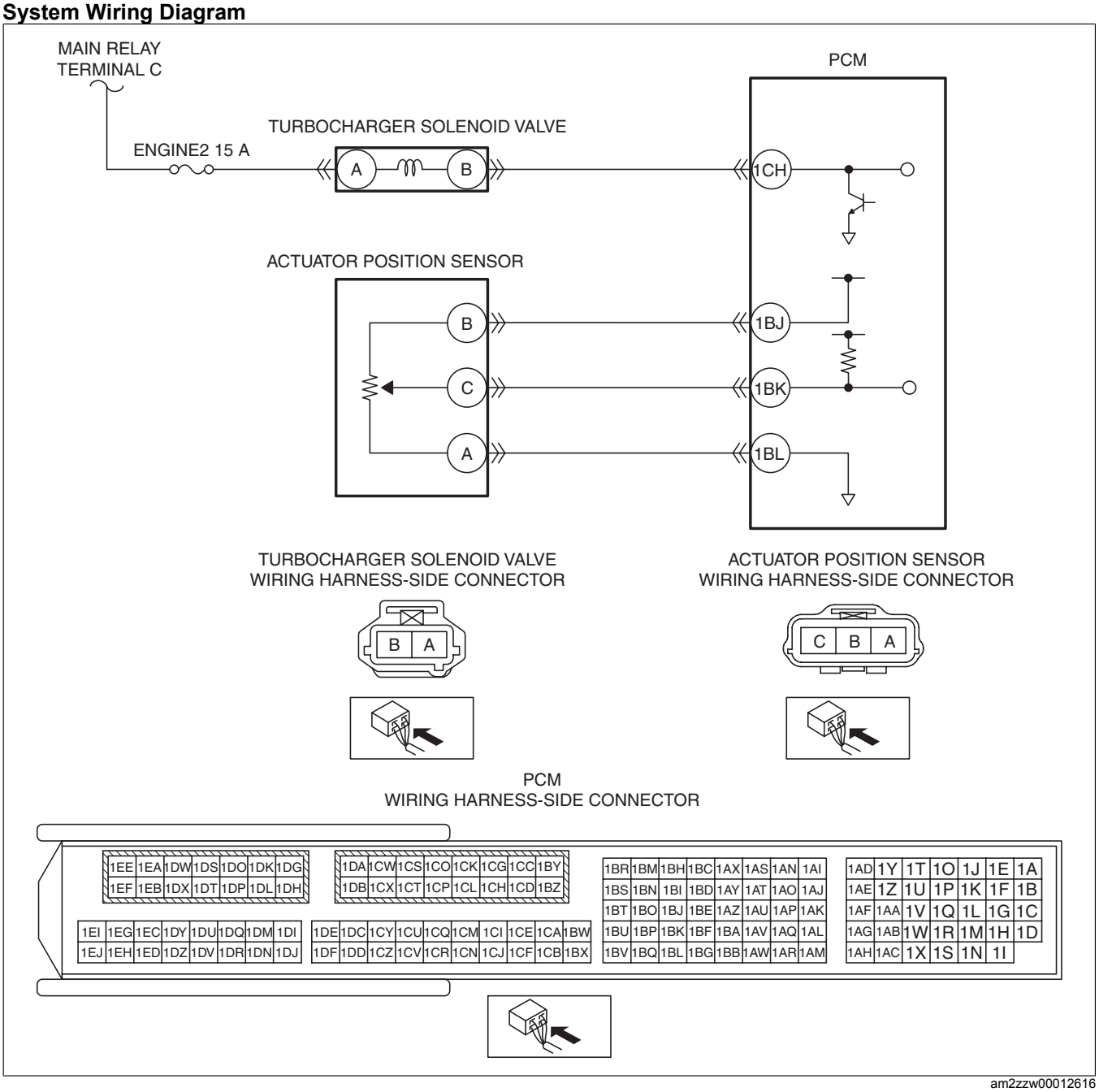
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Note
- To determine the malfunctioning part, proceed with the diagnostics from “Function Inspection Using M-MDS”.

Details On DTCs

DESCRIPTION	Turbocharger underboost condition	
DETECTION CONDITION	Determination conditions	• The difference between the actual intake air pressure and the target intake air pressure less than the specified value for a continuous 7 s when the following conditions are met.
	Preconditions	• 2 s have elapsed with all of the following conditions met when DPF regeneration control is stopped during boost control feedback — Fuel injection amount: **20 mm^3/stoke or more** — Engine speed: **1,400 rpm or more** — Battery voltage: **8 V or more** — Time from engine start: **30 s or more** — Barometric pressure: **70 kPa {0.72 kgf/cm^2, 11 psi} or more** — The following DTCs are not detected: • Fuel pressure sensor: P0192:00, P0193:00 • IAT sensor No.1: P0112:00, P0113:00 • MAP sensor: P0107:00, P0108:00 • BARO sensor: P2228:00, P2229:00 • Exhaust gas pressure sensor No.1: P0471:00, P0472:00, P0473:00 • Intake shutter valve control: P2101:00, P2118:00, P1589:00 • Intake shutter valve position sensor: P0122:00, P0123:00 • Actuator position sensor (turbocharger with variable turbine geometry): P2564:00, P2565:00 • Turbocharger solenoid valve: P0047:00, P0048:00 • Internal PCM malfunction: P062B:00 • Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00
	Drive cycle	• **2**
	Self test type	• CMDTC self test
	Sensor used	• MAP sensor
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Check engine light is illuminated	
POSSIBLE CAUSE	• Erratic signal to PCM — MAP sensor signal malfunction — Input signal part connector or terminals malfunction — Input signal part related circuit malfunction • Exhaust gas leakage from exhaust system • Air suction in intake air system between turbocharger and intake manifold • Vacuum piping or positive pressure piping of turbocharger actuator malfunction • Turbocharger solenoid valve malfunction • Actuator position sensor (turbocharger with variable turbine geometry) malfunction • Turbocharger actuator malfunction • Turbocharger malfunction • PCM malfunction	

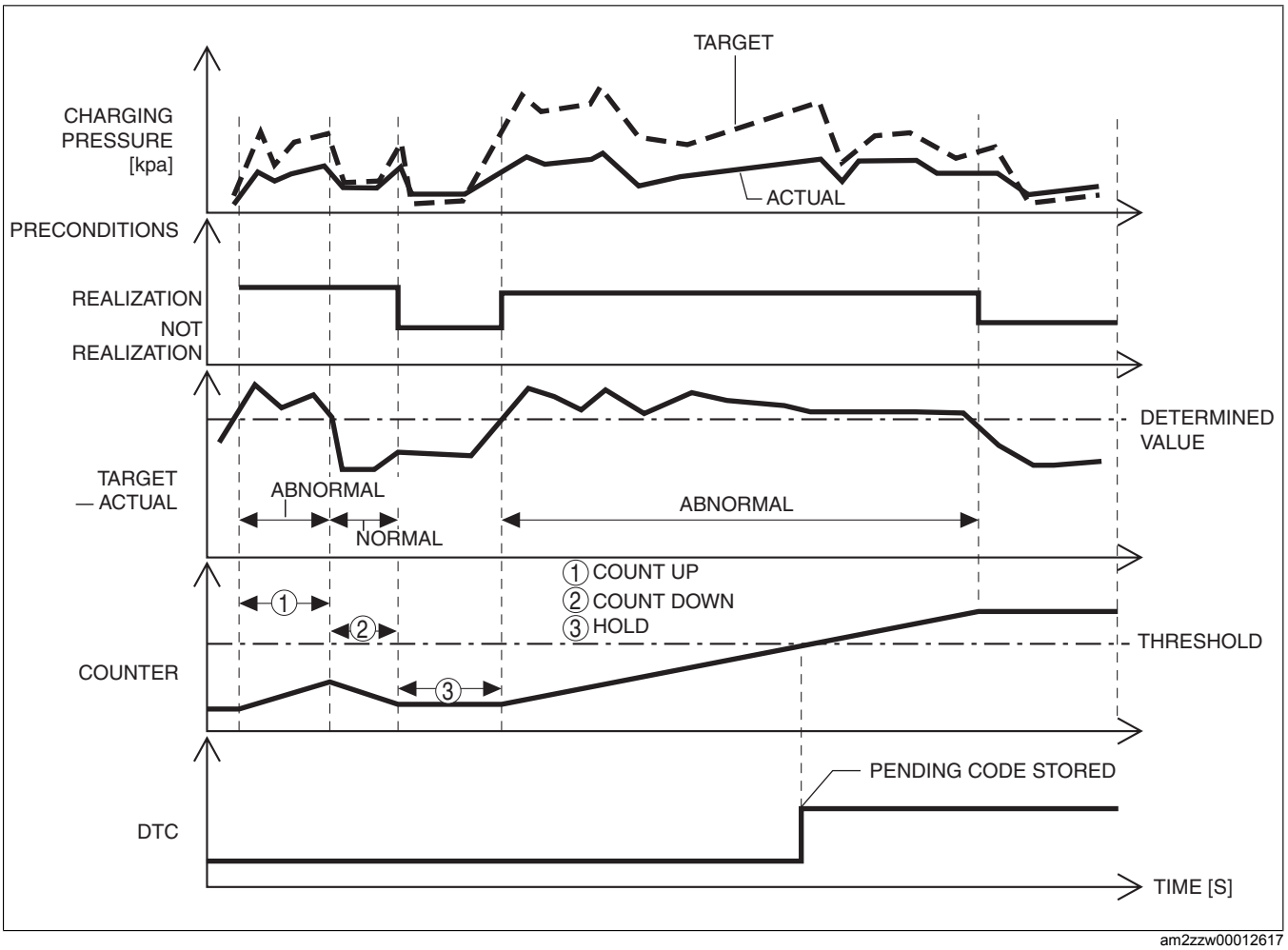
System Wiring Diagram



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Function Explanation (DTC Detection Outline)

- The PCM diagnoses the turbo system performance by comparing the target air charging pressure with the actual.
- The PCM uses the manifold absolute pressure sensor to monitor the intake manifold pressure during boost control as the actual air charging pressure.
- When the monitoring conditions are met during boost control, the PCM compares the target air charging pressure with the actual. If the actual air charging pressure is lower than the specification for the target air charging pressure, the PCM determines an inadequate air charging malfunction.
- The PCM performs diagnosis when each of the preconditions during drive cycle is met. If a malfunction is detected, the malfunction detection counter begins counting up. The value of the malfunction detection counter is maintained while the preconditions are not being met. If the cumulative value of the malfunction detection counter exceeds the threshold during the first drive cycle, a malfunction is determined and a pending code is stored.
- If the PCM determines that the malfunction recurs from the next drive cycle and thereafter, it stores a DTC and turns on the check engine light.



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Repeatability Verification Procedure

Caution

- **When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.**
- **While performing this step, always operate the vehicle in a safe and lawful manner.**

1. Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
   — MAP_DSD
   — MAP
2. Start the engine and warm it up completely.
3. Shift to 2nd gear and rapidly accelerate the vehicle to 40 km/h {25 mph}.
4. Accelerate in 2nd gear from 40 km/h {25 mph} to 55 km/h {35 mph} for 5 s or more.
5. Verify that the difference between MAP_DSD and MAP during acceleration is less than 70 kPa {0.71 kgf/cm^2, 10 psi}.

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
MAP_DSD	Manifold absolute pressure desired value	kPa, Bar, psi	• Displays the boost air pressure desired value
MAP	Manifold absolute pressure	kPa, Bar, psi	• Switch ignition ON (engine off): Approx. 100 kPa {1 bar, 14.5 psi} • Idle: Approx. 80 kPa {0.8 bar, 12 psi}

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P0299:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-D 1.5].)
3	**PURPOSE: RECORD FREEZE FRAME DATA/ SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (misfire related) on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFY IF DIAGNOSTIC RESULT IS AFFECTED BY OTHER RELATED DTCs OCCURRING** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the other PENDING CODE/DTC also present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**PURPOSE: VERIFY IF THERE IS PID ITEM CAUSING DRASTIC CHANGES OF ACCELERATION FLUCTUATION BY INPUT SIGNAL TO PCM** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — MAP_DSD — MAP • Is there any signal that is far out of specification?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.
6	**PURPOSE: VERIFY CONNECTOR CONNECTIONS** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — MAP_DSD — MAP • When the following parts are shaken, does the PID value include a PID item which has changed? — MAP sensor — PCM	Yes	Repair or replace the applicable connector parts. Go to the troubleshooting procedure to perform the procedure from Step 8.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1—3
  — Perform inspection of piping and hoses in intake air system and exhaust system.
- Step 4—7
  — Perform inspection of each separate part.
- Step 8—9
  — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION		ACTION
1	**PURPOSE: INSPECT EXHAUST SYSTEM FOR LEAKAGE** • Visually inspect for exhaust gas leakage from the exhaust system. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 8.
		No	Go to the next step.
2	**PURPOSE: INSPECT INTAKE AIR SYSTEM FOR AIR SUCTION** • Inspect for air leakage at the following: — Around connection of turbocharger and intake manifold **Note** • Engine speed may change when rust penetrating agent is sprayed on the air suction area. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 8.
		No	Go to the next step.
3	**PURPOSE: INSPECT VACUUM PIPING AND POSITIVE PRESSURE PIPING OF TURBOCHARGER ACTUATOR** • Inspect vacuum piping and positive pressure piping of turbocharger actuator. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].) • Is there hose leakage or damage in the vacuum piping and positive pressure piping?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 8.
		No	Go to the next step.
4	**PURPOSE: INSPECT TURBOCHARGER SOLENOID VALVE** • Inspect the turbocharger solenoid valve. (See TURBOCHARGER SOLENOID VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger solenoid valve, then go to Step 8. (See TURBOCHARGER SOLENOID VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**PURPOSE: INSPECT TURBOCHARGER ACTUATOR** • Inspect the turbocharger solenoid valve. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger, then go to Step 8. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**PURPOSE: INSPECT ACTUATOR POSITION SENSOR** • Inspect the actuator position sensor (turbocharger with variable turbine geometry). (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger, then go to Step 8. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
7	**PURPOSE: INSPECT TURBOCHARGER** • Inspect the turbocharger. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger, then go to the next step. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**PURPOSE: PERFORM DTC INSPECTION AND VERIFY IF MALFUNCTIONING PART IS PCM** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
9	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.